

“FROM BALE TO BASKET: FORECASTING SALES IN SECOND-HAND FASHION”



**A machine Learning Approach to Improve
Inventory, Planning and Revenue Forecasts**



DSC-PT-11

GROUP 2

CAPSTONE PROJECT

DSC-PT-11

Authors:

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Kate Maina

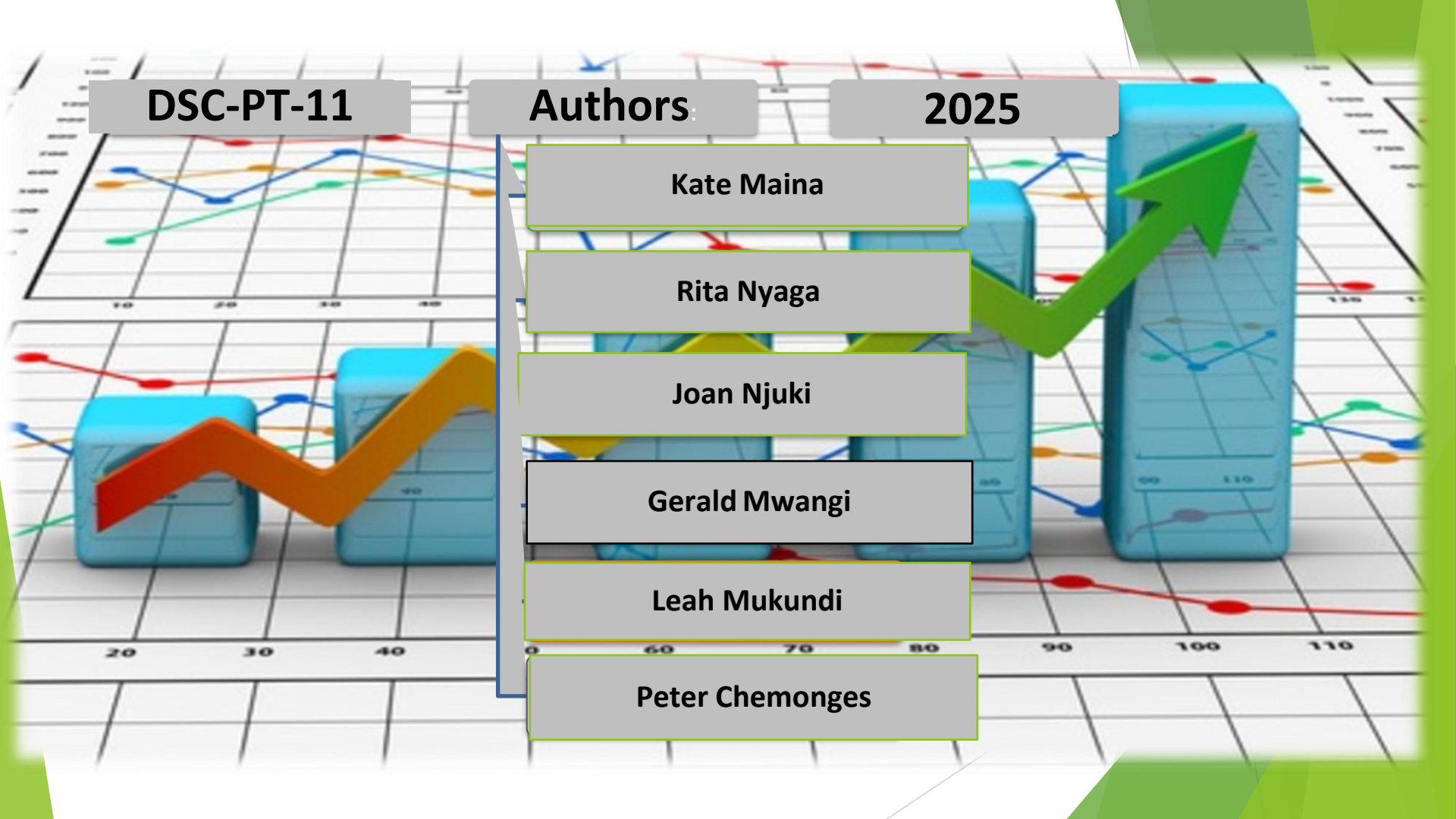
Rita Nyaga

Joan Njuki

Gerald Mwangi

Leah Mukundi

Peter Chemonges



Project Overview

This project develops a data-driven sales forecasting framework for Kenya's second-hand fashion retail sector. By applying statistical, machine learning, and deep learning models, it captures seasonal trends, store-level variations, and product demand patterns. The goal is to replace manual, reactive decision-making with predictive insights that improve inventory planning, reduce stockouts and overstocking, and enhance revenue predictability—ultimately driving operational efficiency and competitiveness in a volatile market

Problem Statement

Kenya's thrift retail sector is highly volatile, with sales performance shaped by irregular import cycles, fluctuating consumer demand, and informal operational practices. Currently, retailers rely on manual, reactive decision-making with no systematic sales forecasting framework in place. This results in frequent stockouts of high-demand items, overstocking of slow-moving products, unreliable revenue projections, inefficient resource allocation, and missed opportunities for growth. A data-driven sales forecasting tool is urgently needed to improve inventory planning, stabilize financial performance, and enable proactive, strategic decision-making in this dynamic market.



Inventory



Inventory



Finance



Finance



Operations

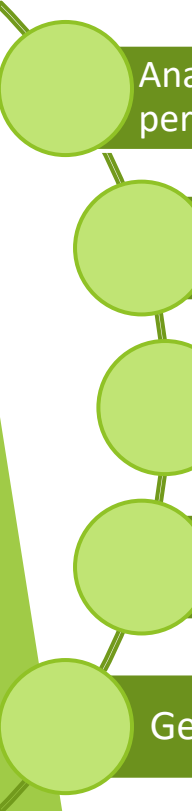


Operations



Growth

Project Objectives



Analyze historical sales data to identify demand patterns, seasonality, and store/product performance.

Develop and compare multiple forecasting models (statistical, ML, and deep learning).

Evaluate model performance using metrics such as MAE, RMSE, and R^2 .

Design a forecasting framework to minimize stockouts and overstocking

Generate actionable insights to improve procurement, pricing, and distribution decisions.

Data Overview

This dataset used contains three years (2022-2024) of historical sales transactions from a Kenyan thrift retail chain. It captures daily sales activity across multiple branches, including product categories, quantities, and revenues.

Key Features:

- **Shop/Branch** – Store location for each transaction
- **Date** – Daily sales records
- **Product/Service** – Item types sold (clothing, accessories, etc.)
- **Transaction Type & No.** – Sales receipts and references
- **Quantity & Sales Value** – Units sold and revenue

Purpose:

The dataset supports sales forecasting by revealing sales patterns, seasonal trends, and store-level variations, enabling better inventory optimization and data-driven decisions.

Dataset Challenges

Inconsistent Product Descriptions-Product names appear in different formats (e.g., Jogging pants vs S/N3, JOGGING PANTS), making grouping unreliable.

Missing or Blank Values- Some fields contain blanks or missing values; wholesale lines lack detail

Transaction-Level Granularity- Highly granular (one line per product per receipt); forecasting on raw rows is noisy

Lack of Product Hierarchy-No structured hierarchy (Category → Sub-category → SKU), complicating analysis.

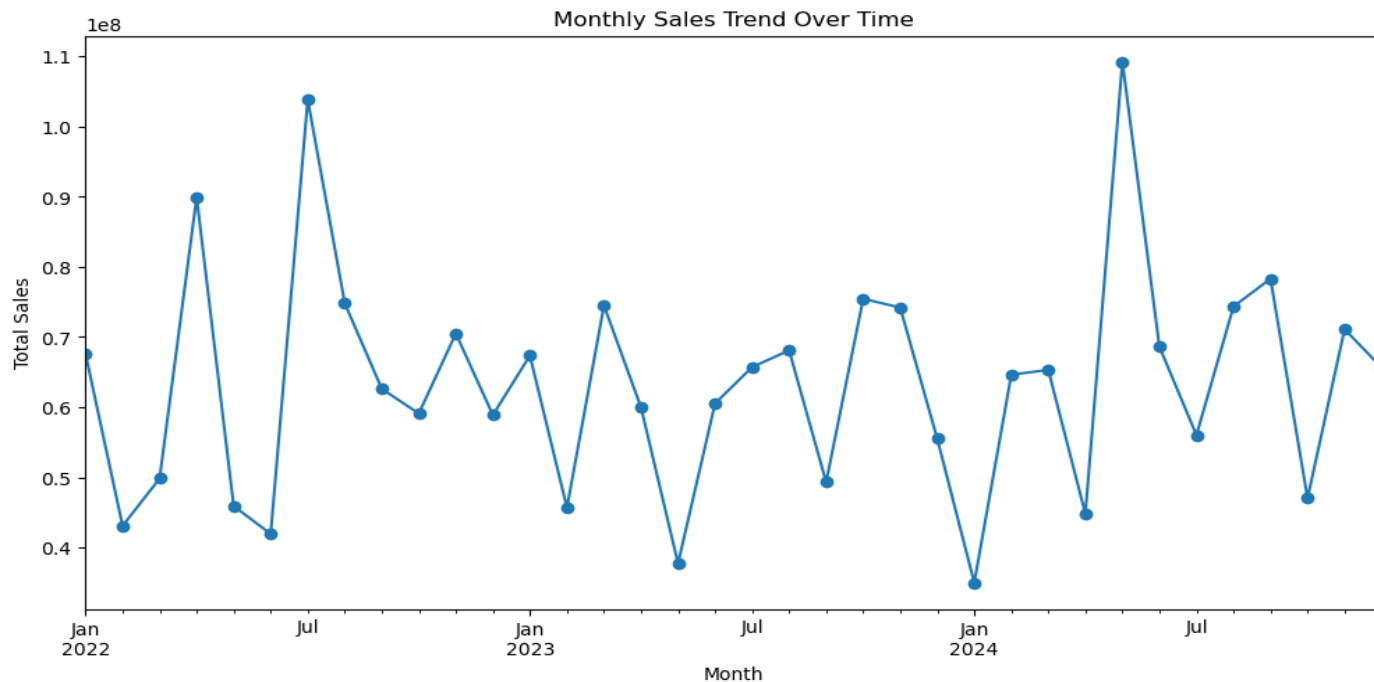
Volatility in Sales-Sales depend on bale arrivals (supply-driven) more than customer demand, introducing randomness.

Intermittent & Sparse Demand -Some products show long zero-sales periods followed by spikes. Classical models struggle.

Cold-Start Products-New product types appear later with no prior history, making forecasting difficult

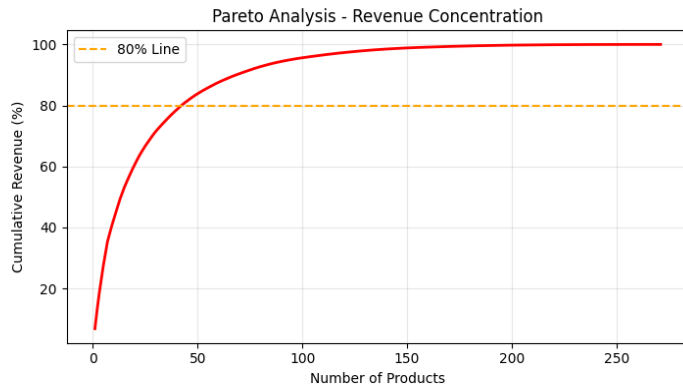
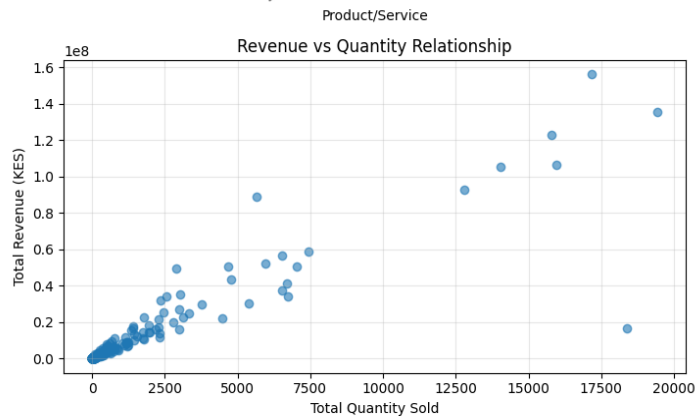
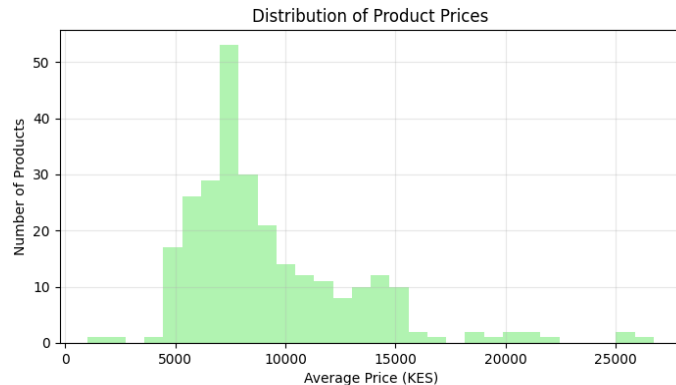
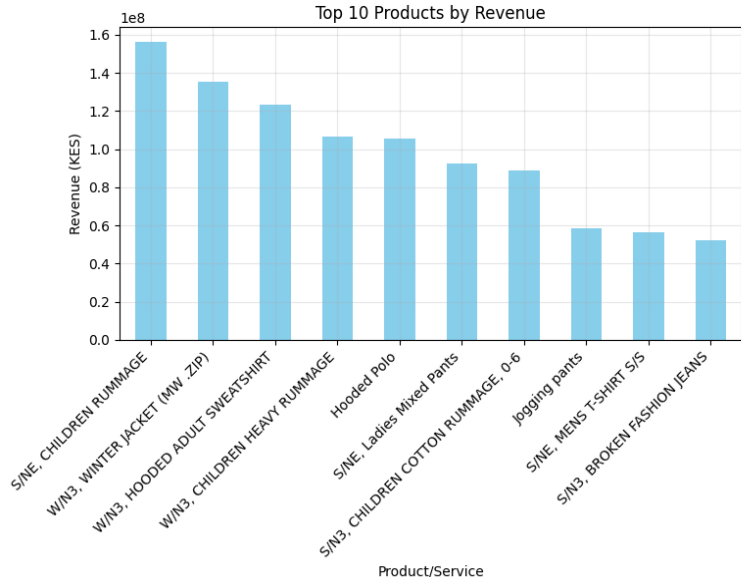
Exploratory Data Analysis

1.Sales Trend Over Time (2022-2024)



Insight: “Sales are highly volatile with sharp peaks (seasonal or event-driven) and deep troughs, underscoring the need for forecasting.”

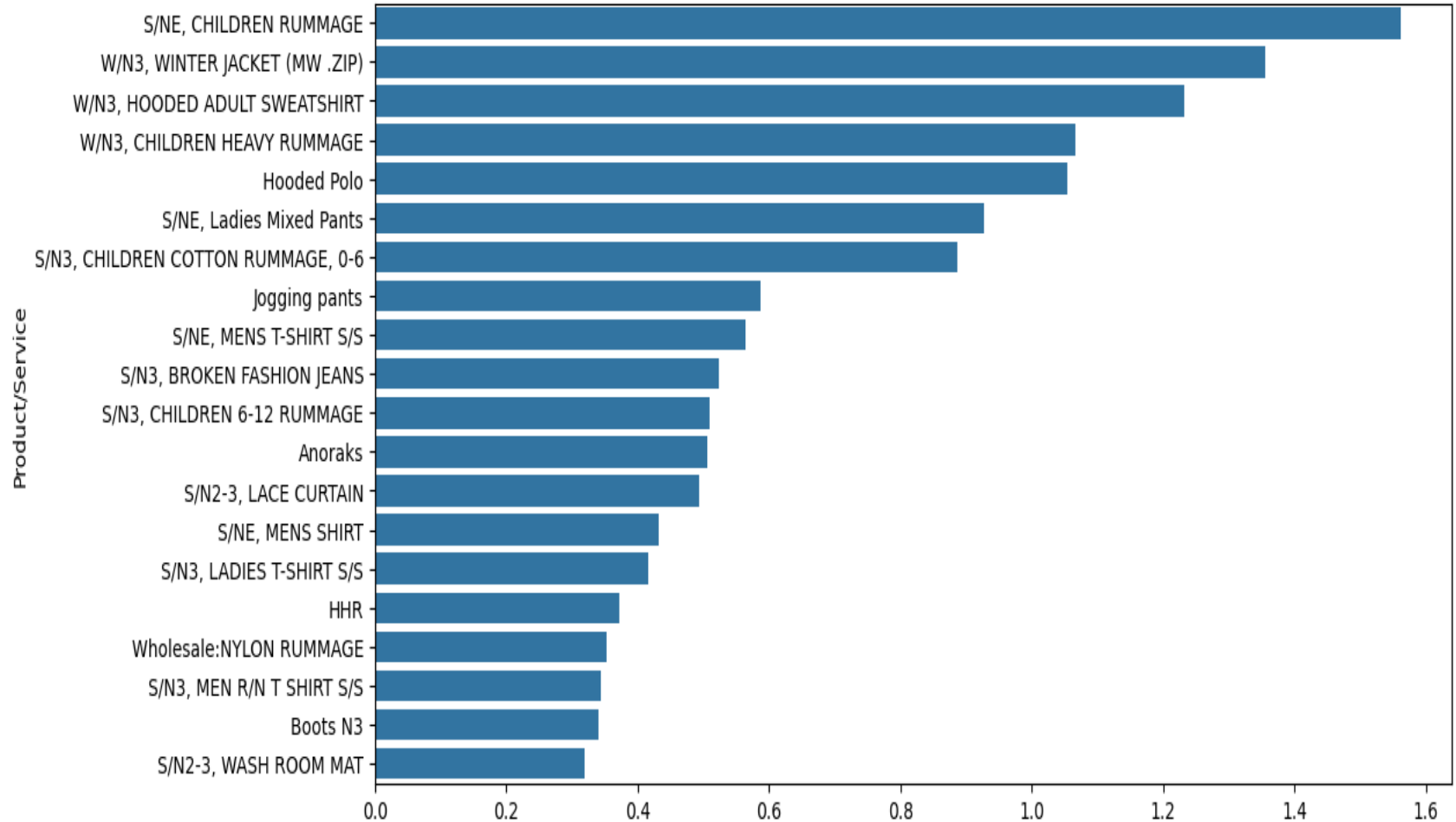
2.Product Revenue Analysis



“Sales are heavily concentrated in a few top products—highlighting the need to diversify mid-tier SKUs and strengthen supply for high performers to stabilize revenue growth..”

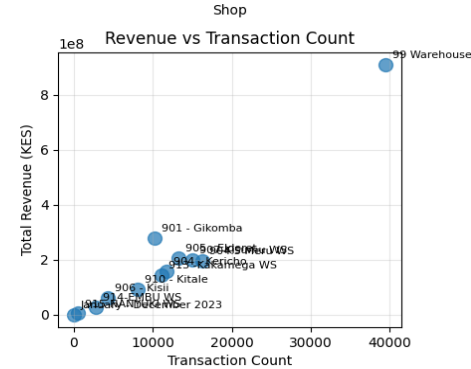
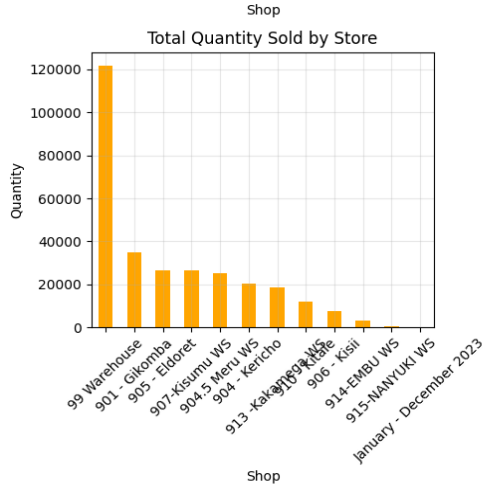
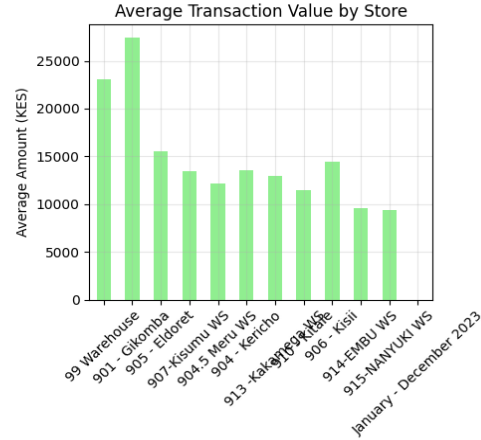
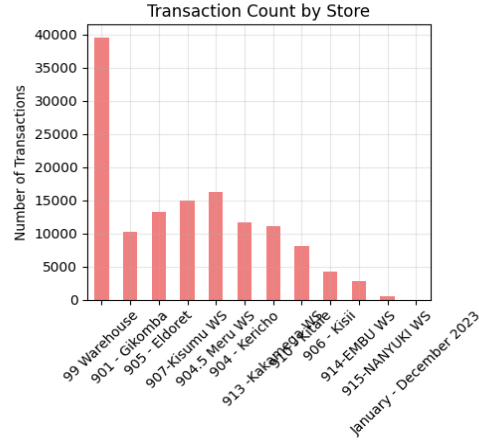
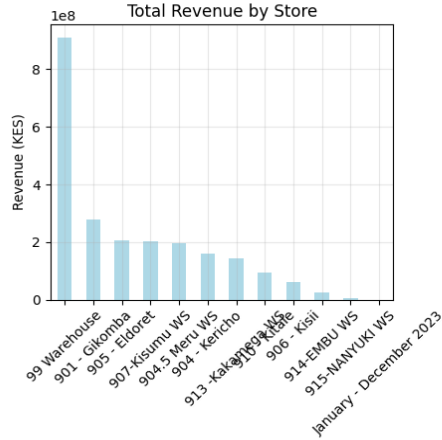
3.Product/Category Contribution to Sales-Amount

Top 20 Products by Total Sales Amount



“Children’s rummage and winter jackets dominate revenue, showing that a small set of products drives most sales value”.

Store Performance analysis

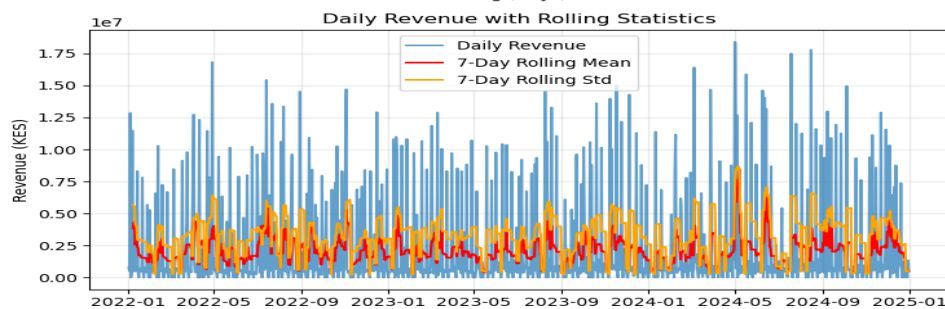
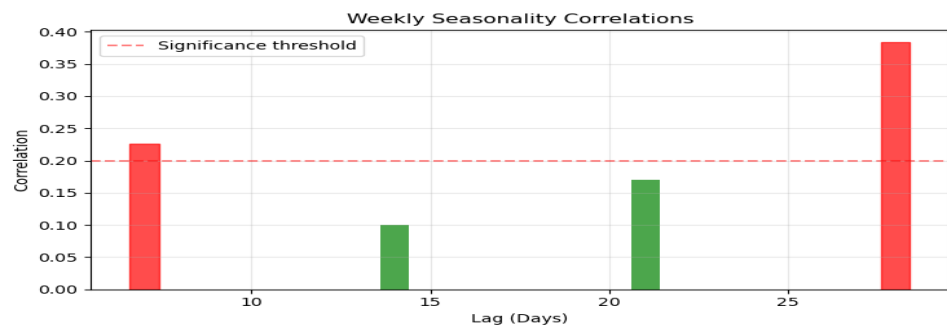
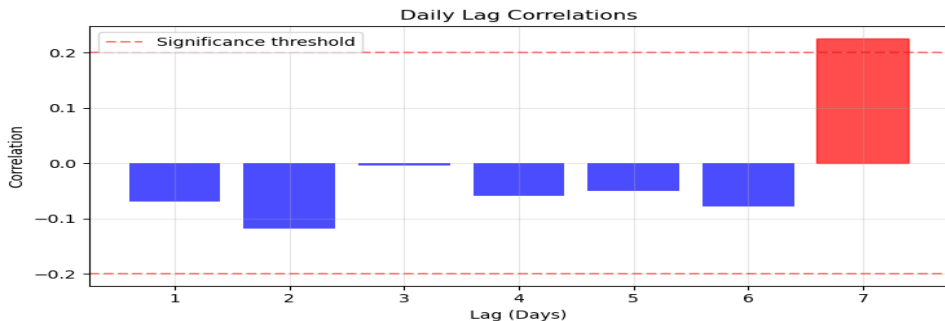
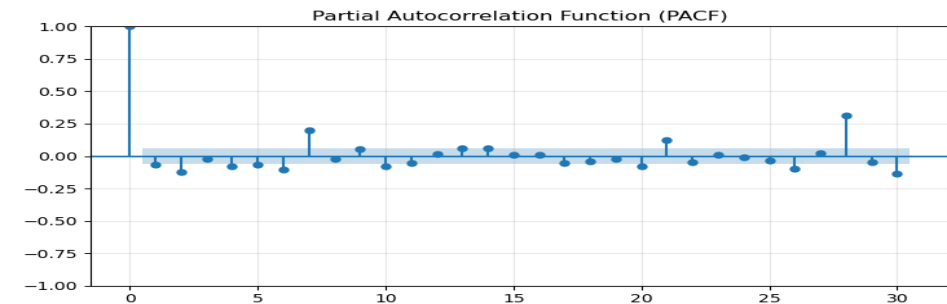
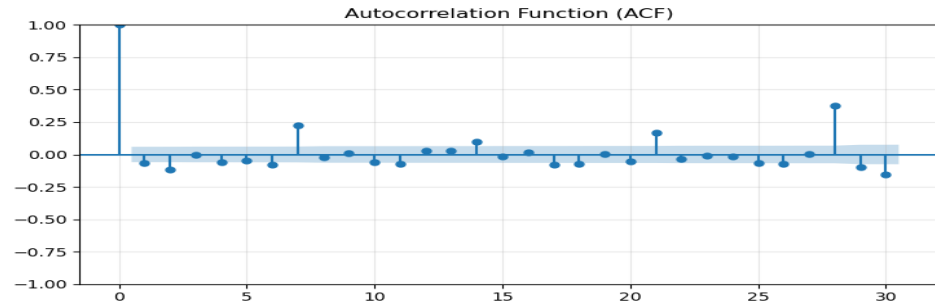
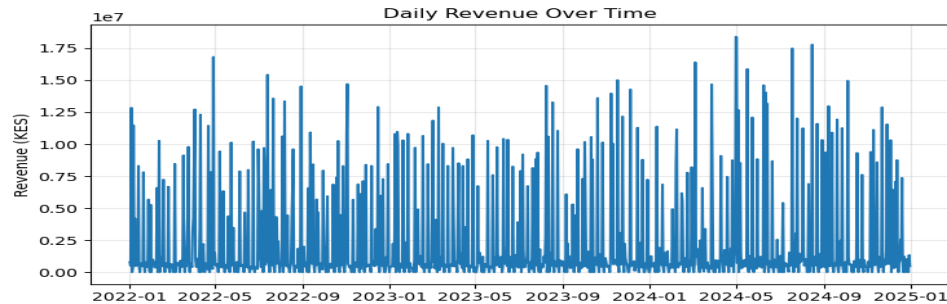


“Current revenue plateau at KES 378M presents both a ceiling requiring breakthrough strategies and a stable foundation for scaling through innovation and market expansion”.

Store Performance

#	Store	Revenue	Transactions	Avg Revenue Per Transaction
1	99 Warehouse	KES 909,623,45.65	39,507	KES 23,024,36
2	901 - Gikomba	KES 279,445,753.53	10,192	KES 27,418,15
3	905 - Eldoret	KES 206,367,245.02	13,296	KES 15,521.00
4	907 - Kisumu WS	KES 201,156,576.25	14,979	KES 13,429.24
5	904.5 Meru WS	KES 196,598,275.67	16,221	KES 12,119.98
6	904 - Kericho	KES 158,551,765.87	11,683	KES 13,571.15
7	913 - Kakamega WS	KES 143,110,559.21	11,055	KES 12,945.32
8	910 - Kitale	KES 93,312,713.04	8,118	KES 11,494.54
9	906 - Kisii	KES 26,687,714.39*	4,279	KES 14,460.83
10	914- Embu WS	KES 4,965,551.62	2,789	KES 9,568.92
11	915 - Nanyuki WS	KES 4,965,551,62	528	KES 9,404.45

Time Series and Lag Analysis



Time Series and Lag Analysis- insights

 **High Revenue Volatility:** Daily sales fluctuate irregularly, showing unstable purchasing patterns.

 **Weak Short-Term Correlation:** ACF/PACF indicate that previous days' sales have little predictive influence on current sales.

 **Emerging Weekly Pattern:** A noticeable spike at lag 7 suggests possible weekly buying or restocking cycles.

 **Non-Stationary Series:** Both mean and variance vary over time — differencing or scaling is needed for stable forecasting.

• **Irregular Seasonality:** Seasonal cycles are inconsistent, reflecting unpredictable customer behavior and irregular bale arrivals.

Modelling

Why Different Models?

- **Baselines** → simple benchmarks.
- **Statistical** → capture trend & seasonality.
- **Machine Learning** → flexible with features.
- **Deep Learning** → handle complex, non-linear patterns.

Goal: Find the most accurate and practical sales forecasting model.

Modeling Approaches



Baseline Models

- Naive Forecast
- Moving Average



Statistical Models

- Prophet
- SARIMA



Machine Learning Models

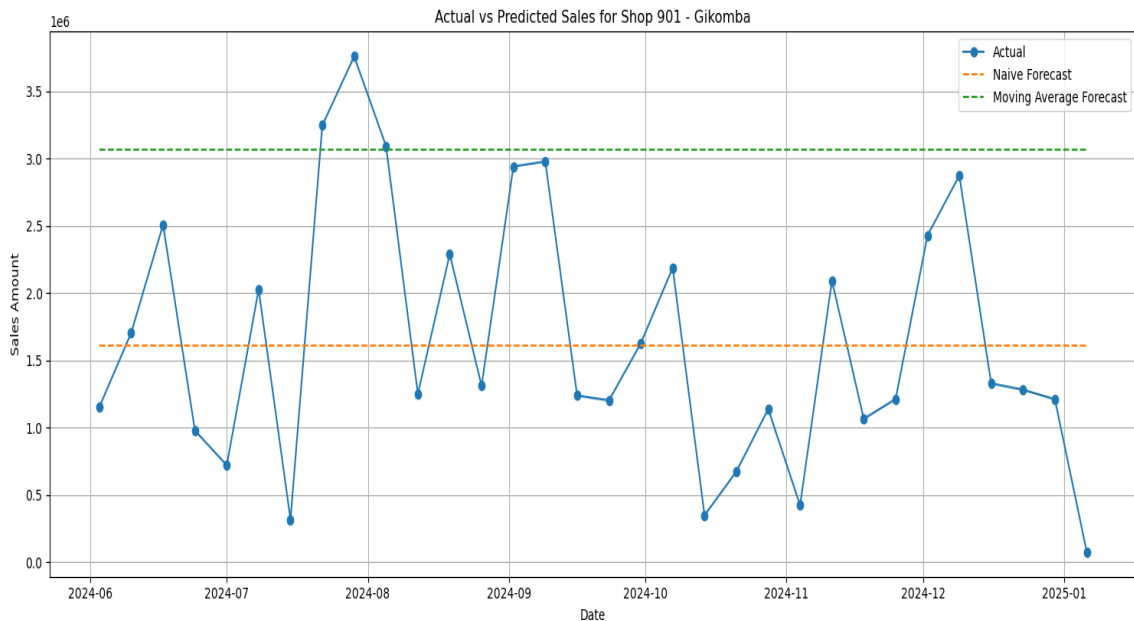
- Random Forest
- XGBoost
- Support Vector Regression (SVR)



Deep Learning Models

- LSTM

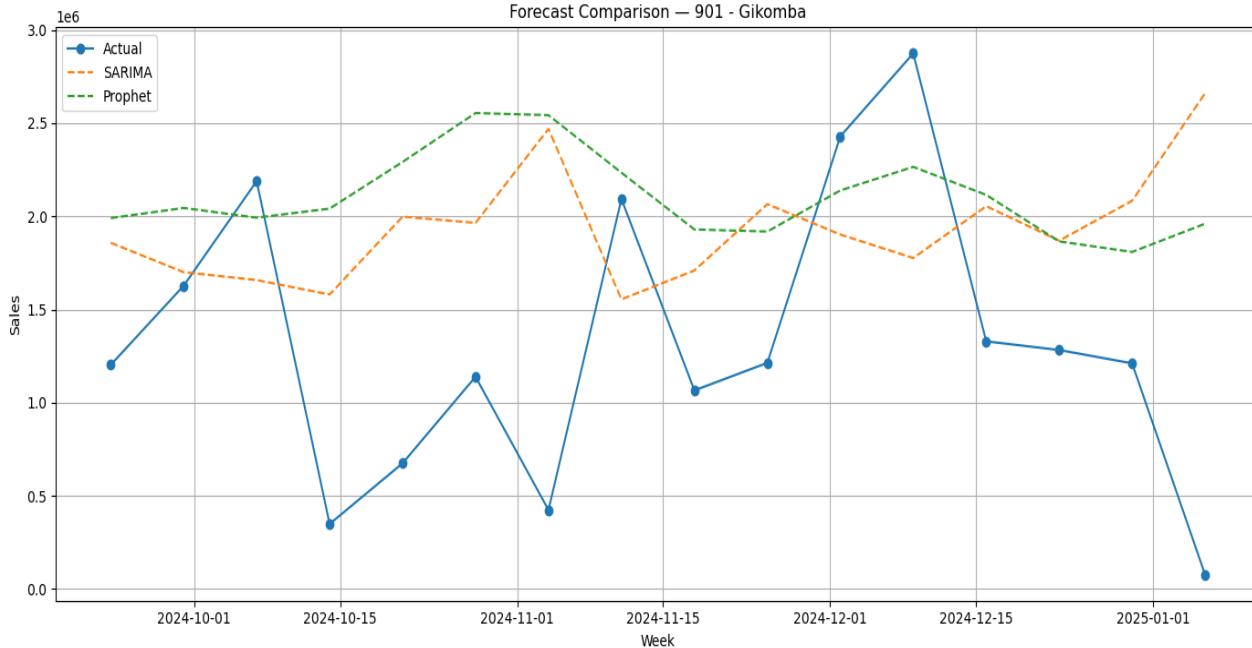
1. Baseline Models



Model	RMSE	MAE	R-squared
Naïve Forecast	1,380,698	622,077	0.58
Moving Average(4)	2,341,833	1,088,732	-0.20

Baselines provide a starting point, but their high errors confirm the need for advanced forecasting models

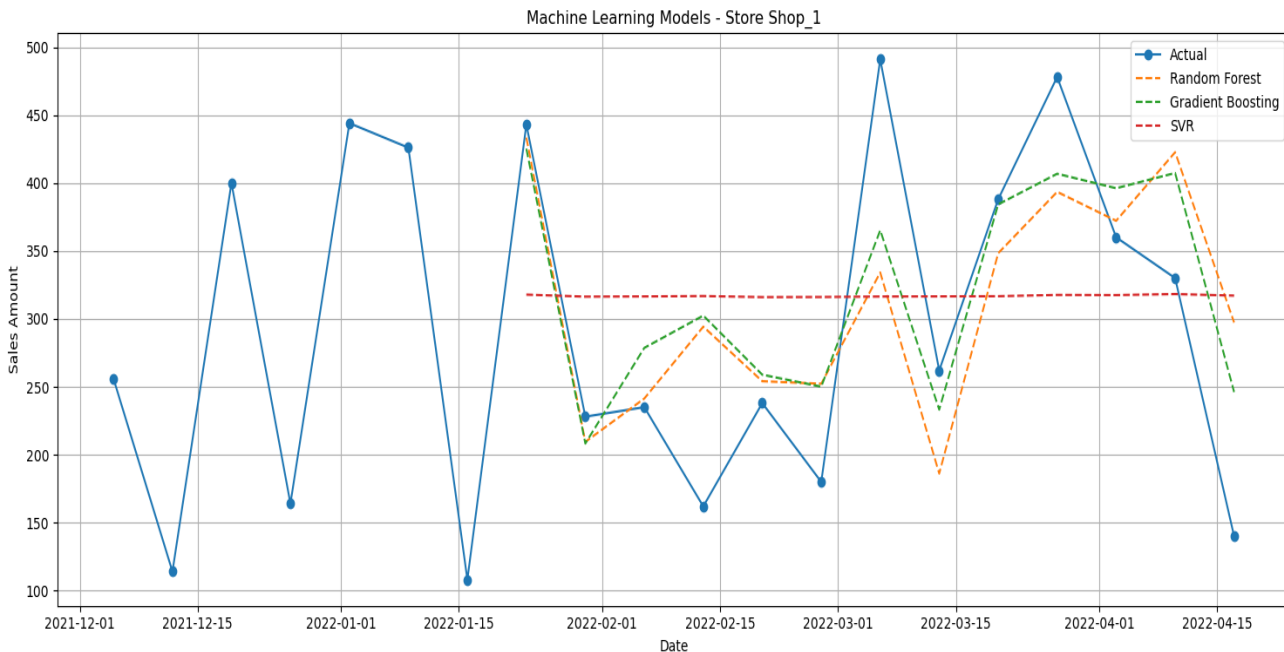
2. Statistical Models



Model	RMSE	MAE	R-squared
SARIMA	1,370,642	675,678	0.54
Prophet	2,091,129	974,897	-0.065

SARIMA outperformed Prophet, with lower RMSE/MAE and a positive R^2 . Prophet struggled to capture volatility, producing high errors and a negative R^2 . Statistical models add value beyond baselines, but still limited for highly irregular sales patterns

2. Machine Learning Models



Model	RMSE	MAE	R-squared
Random Forest	92.9	76.3	0.33
Gradient Boosting	83.6	65.6	0.46
SVR	108.3	91.2	0.09

Gradient Boosting outperformed other ML models, proving the effectiveness of tree-based ensemble methods for sales forecasting.

2. Deep Learning Model

Model	RMSE	MAE	R-squared
LSTM	1,327,272	640,973	0.56

“With an R^2 of 0.56, the LSTM model demonstrated moderate predictive strength, confirming the viability of deep learning for retail demand forecasting.”

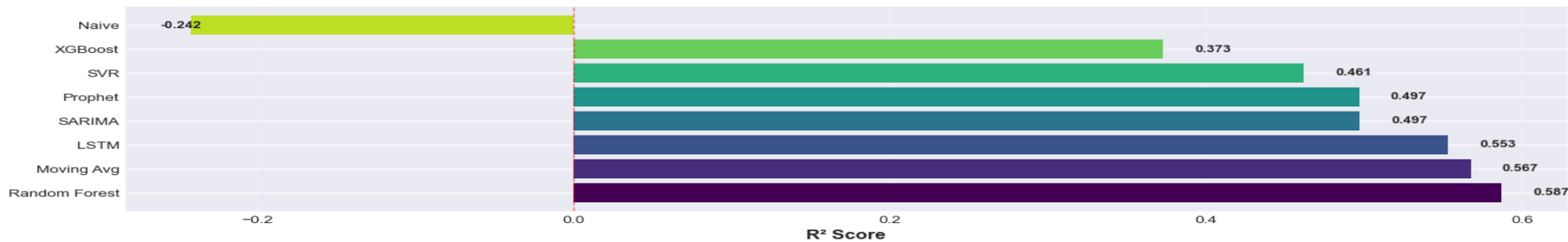
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Model Selection Justification

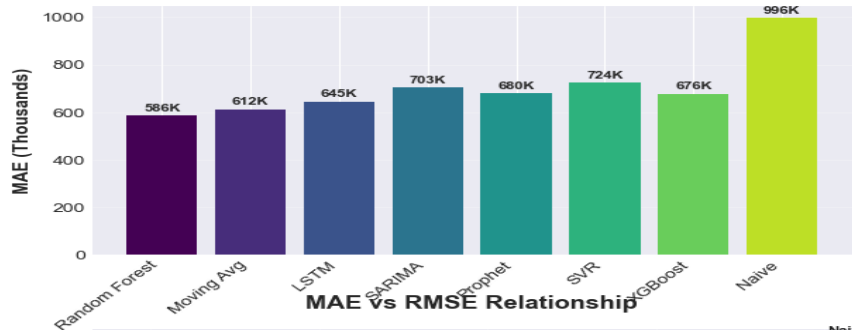
Model	Dataset Type	MAE (KES)	RMSE (KES)	R ²	Remarks
Random Forest (FE, pooled)	Feature-Engineered	585,788	1,279,279	0.587	Best overall performer — chosen for deployment
Gradient Boosting (FE, pooled)	Feature-Engineered	604,400	1,372,581	0.524	Good accuracy but less stable than RF
LSTM (pooled + calendar + shop)	Feature-Engineered	644,868	1,329,152	0.554	Captures trends but slightly less accurate
Moving Average (4)	Plain Dataset	612,119	1,309,287	0.567	Simple baseline model
SARIMA (1,1,1)x(1,1,1,52)	Plain Dataset	703,284	1,411,438	0.497	Handles seasonality, weaker accuracy
Prophet (weekly + yearly)	Plain Dataset	679,648	1,411,875	0.497	Consistent but less adaptive
SVR (FE, pooled)	Feature-Engineered	723,997	1,460,451	0.461	Overfits and generalizes poorly
Naïve (last value)	Plain Dataset	995,755	2,218,043	-0.242	Benchmark baseline — lowest performance

Model Performance Comparison Dashboard

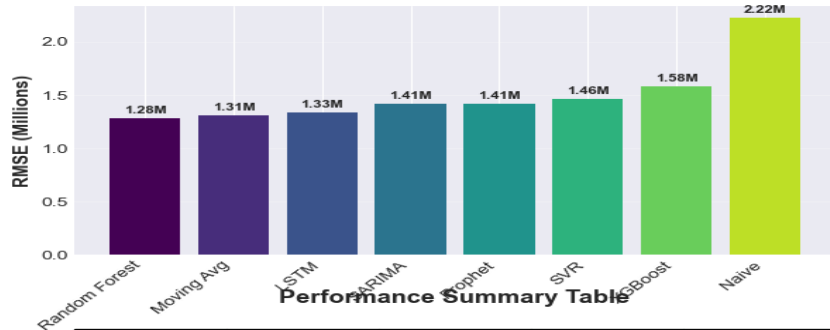
R² Score Comparison (Higher is Better)



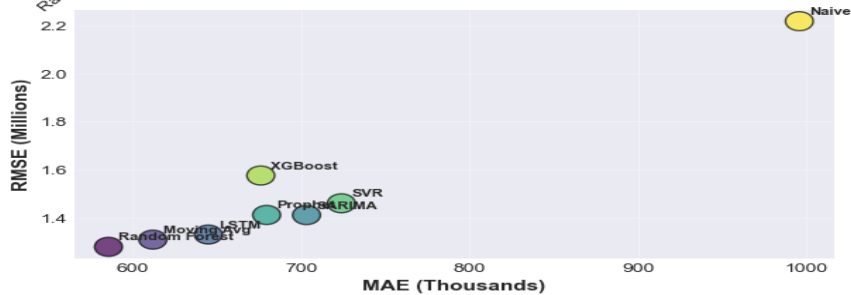
Mean Absolute Error (Lower is Better)



Root Mean Squared Error (Lower is Better)



MAE vs RMSE Relationship



Performance Summary Table

Rank	Model	MAE	RMSE	R ²
1	Random Forest	585.8K	1.28M	0.587
2	Moving Avg	612.1K	1.31M	0.567
3	LSTM	645.0K	1.33M	0.553
4	SARIMA	703.3K	1.41M	0.497
5	Prophet	679.6K	1.41M	0.497
6	SVR	724.0K	1.46M	0.461
7	XGBoost	676.2K	1.58M	0.373
8	Naive	995.8K	2.22M	-0.242

Model Deployment

1-Model Selection & Tuning

The best-performing model (Random Forest) was selected based on RMSE, MAE, and R^2 scores.

2-Model Serialization

The trained model was exported using joblib for easy reusability.

3-API Integration

A REST API was built (Flask) to serve forecasts to business systems.

4-Dashboard Deployment

Streamlit dashboard for real-time visualization of predicted sales trends.

5-Business Utilization

Store managers and finance teams access sales forecasts for inventory and budget decisions.

Conclusion

This project proved that predictive modeling can revolutionize how Kenyan thrift retailers plan, stock, and sell. By turning raw sales data into foresight, we built a forecasting system that brings clarity to chaos.

✦ Key Takeaways

- Random Forest wins the race: It outperformed traditional models in capturing complex, multi-store sales patterns.**
- Smarter every season: Retraining the model and adding variables like holidays or promotions will keep forecasts razor-sharp.**
- From guesswork to growth: Data-driven planning now replaces intuition — cutting waste, improving cash flow, and driving strategic expansion.**

Recommendations

Deploy Random Forest as the core forecasting model and automate regular retraining cycles.

Integrate forecasts into a BI dashboard (e.g., Tableau) for real-time store insights.

Enhance model accuracy by adding external factors like promotions ,discounts,and local events

Explore hybrid approaches combining Random Forest and LSTM for stronger forecasts

Adopt forecast-driven decision-making across inventory, staffing, and procurement.



Thank You

Thank you for your
time and
attention.

We appreciate
your support and
interest in our
project.

— *The Team*

thank
you